AI-Based Student Presence Detection in University Surveillance Videos via Face Recognition

CAPSTONE PROJECT REPORT

Submitted by

Buchupalli Lalithadithya Reddy– 99220041142

Chamanthula Shiva Kumar – 99220041150

Bairaju Naveen Chandra – 99220041118

Chandragiri Mohammad Ashraf – 99220041151

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KRISHNANKOIL 626 126

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DECLARATION

We affirm that the project work titled **"AI-Based Student Presence Detection in University Surveillance Videos via Face Recognition"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

B. Lalithadithya Reddy

99220041142

B. Naveen Chandra

99220041118

C. Shiva Kumar

99220041150

C. Mohammad Ashraf

99220041151

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Signature of supervisor

Dr. M. K. Nagarajan

Associate Professor

Department of Computer Science and Engineering

KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade

BONAFIDE CERTIFICATE

Certified that this project report **"AI-Based Student Presence Detection in University Surveillance Videos via Face Recognition"** is the bonafide work of "**B. Lalithadithya Reddy (99220041142), B. Naveen Chandra (99220041118), C. Shiva Kumar (99220041150), C. Mohammad Ashraf (99220041151)"** who carried out the project work under my supervision.

Dr. M. K. Nagarajan
SUPERVISOR
Associate Professor
Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil 626126
Virudhunagar District.

Dr. R. Raja Subramanian
HEAD OF THE DEPARTMENT
Associate Professor & Head
Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil 626126
Virudhunagar District.

Submitted for the Project Viva-voce examination held on

Internal Examiner **External Examiner**

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COMPUTER SCIENCE AND ENGINEERING

PROJECT SUMMARY

Project Title	**AI-Based Student Presence Detection in University Surveillance Videos via Face Recognition**	
Project Team Members (Name with Register No)	B. Lalithadithya Reddy - 99220041142 C. Shiva Kumar - 99220041150 B. Naveen Chandra - 99220041118 C. Mohammad Ashraf - 99220041151	
Guide Name/Designation	Dr. M. K. Nagarajan	
Program Concentration Area	Face Recognition	
Technical Requirements	Computer Vision, Image Processing, Python Programming, Flask Web Framework, OpenCV Library, FaceNet Model, Database Management System (MySQL/SQLite), HTML, CSS, JavaScript, Cloud/Local Storage.	
Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	The system is designed using cost-effective software tools such as Python, OpenCV, and Flask which are open-source. Hardware requirements like CCTV cameras and storage systems are selected based on affordability and scalability to reduce overall implementation cost.	✓
Environmental	The project minimizes environmental impact by using digital surveillance instead of manual monitoring, reducing paper usage and physical security resources. Energy consumption is limited to CCTV and server operation.	✓
Social	The system enhances public safety and crime prevention by enabling real-time face recognition in campuses, offices, and public places. It supports law enforcement and institutional monitoring.	✓

Ethical	Facial data collection and storage follow ethical guidelines ensuring user consent, privacy protection, and secure handling of biometric information to prevent misuse.	✓
Health and Safety	The system improves safety by monitoring restricted areas, detecting unauthorized persons, and assisting in emergency response without physical human risk.	✓
Manufacturability	The project is software-centric and can be deployed on existing CCTV infrastructure without complex manufacturing processes, making implementation feasible and scalable.	✓
Sustainability	The solution promotes long-term sustainability by enabling automated surveillance, reducing manpower requirements, and supporting smart security infrastructure development.	✓

REALISTIC CONSTRAINTS:

Economic Constraint:

The development and deployment of the Smart Face Recognition Surveillance System involve cost considerations related to hardware infrastructure such as CCTV cameras, storage servers, and computational resources. To address economic constraints, the proposed system utilizes open-source software technologies including Python, OpenCV, and Flask, which significantly reduce software licensing costs. The system is also designed to operate on existing surveillance infrastructure, minimizing additional investment and making it economically feasible for institutions and organizations.

Environmental Constraint:

The environmental impact of the system is minimal as it primarily operates through digital processing and automated monitoring. Unlike traditional surveillance methods that require extensive manual documentation and physical resource utilization, this system reduces paper usage and administrative overhead. Energy consumption is limited to camera operation and server processing, and efficient data storage practices are adopted to optimize power utilization.

Social Constraint:

From a societal perspective, the system contributes positively by enhancing public security, crime prevention, and institutional safety. It can be deployed in educational institutions, workplaces, transportation hubs, and public areas to monitor unauthorized access and suspicious activities. However, societal acceptance depends on transparent usage policies and awareness regarding its benefits and limitations.

ABSTRACT

In today's world, security and surveillance have become essential requirements for institutions, organizations, and public environments. Traditional Closed Circuit Television(CCTV) monitoring systems rely heavily on manual observation, which is time-consuming and prone to human error. To overcome these limitations, this project proposes a Smart Face Recognition System in Closed Circuit Television (CCTV) Footage that automates human identification using Artificial Intelligence and Computer Vision technologies. The system is designed to detect and recognize human faces from live CCTV feeds as well as recorded video footage. It uses advanced deep learning algorithms, particularly the FaceNet model integrated with OpenCV libraries, to extract facial features and perform accurate recognition. The application allows users to upload surveillance videos, automatically extracts frames, detects faces present in each frame, and compares them with a pre-registered facial database.

In addition to recognition, the system displays detailed information of identified individuals such as name, registration number, department, and contact details through an interactive web interface developed using the Flask framework. If a face is not found in the database, the system labels it as "Unknown," thereby assisting in identifying unauthorized persons. This feature enhances institutional security and monitoring efficiency.The proposed solution is cost-effective as it utilizes open-source technologies and can be integrated with existing CCTV infrastructure without requiring major hardware modifications. Experimental results demonstrate that the system achieves high accuracy in face detection and recognition under controlled conditions.Overall, this project provides an intelligent, automated, and scalable surveillance solution that reduces manual effort, improves real-time monitoring, and strengthens safety measures in smart environments. Future enhancements may include real-time alert systems, cloud deployment, and integration with law enforcement databases.

Keywords: Face Recognition, Closed Circuit Television (CCTV) Surveillance, Deep Learning, Computer Vision, FaceNet, OpenCV, Image Processing, Human Identification, Video Frame Extraction, Smart Security System, Flask Web Application, Artificial Intelligence

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LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1	211CSE1402	Python Programming
2	213CSE1301	Introduction to Artificial Intelligence and Machine Learning
3	212CSE2304	Machine Learning
4	212CSE2305	Database Management Systems
5	212CSE2303	Software Engineering

LIST OF ABBREVIATIONS USED

S.NO	ABBREVIATION	FULL FORM
1	CCTV	Closed Circuit Television
2	AI	Artificial Intelligence
3	ML	Machine Learning
4	CNN	Convolutional Neural Network
5	DNN	Deep Neural Network
6	FaceNet	Face Embedding Network
7	OpenCV	Open Source Computer Vision
8	GUI	Graphical User Interface
9	UI	User Interface
10	DB	Database
11	SQL	Structured Query Language
12	CPU	Central Processing Unit
13	GPU	Graphics Processing Unit
14	FPS	Frames Per Second
15	ROI	Region of Interest
16	RGB	Red Green Blue
17	JPEG	Joint Photographic Experts Group
18	PNG	Portable Network Graphics
19	MP4	MPEG-4
20	AVI	Audio Video Interleave
21	RTSP	Real Time Streaming Protocol

22	HTML	HyperText Markup Language
23	CSS	Cascading Style Sheets
24	JS	JavaScript
25	UAT	User Acceptance Testing
26	DFD	Data Flow Diagram
27	ER	Entity Relationship
28	FAR	False Acceptance Rate
29	FRR	False Rejection Rate
30	UVPAS	University Video Processing and Analysis System

CHAPTER – I

INTRODUCTION

1.1 Background and Motivation

The rapid advancement of digital technologies and smart infrastructure has significantly transformed the way security and surveillance are managed across institutions and public environments. Closed Circuit Television (CCTV) systems are now widely deployed in educational institutions, corporate organizations, transportation hubs, residential complexes, and public spaces to monitor activities and ensure safety. These systems continuously record video footage, which can be reviewed when incidents occur. However, traditional CCTV surveillance still relies heavily on manual monitoring, where security personnel must observe multiple video feeds simultaneously.

Manual monitoring is not only time-consuming but also inefficient, as human operators are prone to fatigue, distraction, and delayed response. Identifying a specific individual from hours of recorded footage is a complex and labor-intensive process. As surveillance data continues to grow exponentially, the need for automated and intelligent monitoring solutions has become increasingly critical.

Artificial Intelligence (AI) and Computer Vision technologies have opened new possibilities in transforming conventional surveillance into smart security systems. Face recognition, a biometric authentication technique, enables automatic identification of individuals by analyzing unique facial features. By integrating face recognition with CCTV footage, surveillance systems can move beyond passive recording to active identification and tracking.

The motivation behind this project is to develop an intelligent CCTV surveillance system capable of automatically detecting and recognizing human faces from video footage. Using deep learning models such as FaceNet along with OpenCV image processing techniques, the system can identify registered individuals and flag unknown persons. This enhances institutional security, reduces manual effort, and enables faster incident investigation. The project aims to bridge the gap between traditional video surveillance and AI-driven smart monitoring solutions.

1.2 Problem Statement

Although CCTV cameras are extensively installed for security purposes, most existing systems function only as recording devices rather than intelligent monitoring platforms. Security personnel must manually scan recorded footage to identify suspects or verify identities, which is inefficient and prone to human error. Moreover, locating a particular individual in large volumes of surveillance data is extremely challenging.

The absence of automated identification mechanisms leads to several issues:

- Delay in identifying unauthorized persons
- Difficulty in tracking suspects across multiple cameras
- Increased dependency on human monitoring
- Inefficient incident investigation

- Limited utilization of recorded surveillance data

Therefore, the core problem is the lack of an automated face recognition mechanism integrated with CCTV systems. There is a need for a smart surveillance solution that can detect faces, recognize individuals using stored datasets, and provide identity details without requiring continuous human supervision.

1.3 Objectives of the Project

The primary aim of this project is to design and develop an AI-based CCTV Face Recognition System that enhances surveillance efficiency and security. The key objectives include:

- Developing a smart surveillance system capable of detecting faces from CCTV footage.
- Implementing deep learning–based face recognition using FaceNet.
- Extracting frames from recorded surveillance videos for analysis.
- Comparing detected faces with pre-registered facial datasets.
- Displaying personal details of recognized individuals.
- Labeling unidentified persons as "Unknown."
- Creating a web-based interface for monitoring and management.
- Reducing manual surveillance effort through automation.
- Improving institutional security and access monitoring.

1.4 Scope of the Project

The proposed system provides an intelligent digital surveillance platform for identifying individuals through CCTV footage. Its scope includes:

- Uploading recorded CCTV videos for analysis.
- Extracting frames automatically from video streams.
- Detecting human faces using computer vision techniques.
- Recognizing faces through deep learning embeddings.
- Matching faces with stored datasets of registered persons.
- Displaying identity details such as name, register number, department, and contact information.
- Marking unknown individuals when no match is found.
- Providing a user-friendly web interface for administrators.

The project is primarily focused on post-event video analysis and recognition, not continuous real-time city surveillance. However, the system architecture can be extended in the future to include:

- Live CCTV face recognition

- Criminal database integration
- Real-time alerts and notifications
- Cloud-based surveillance storage
- Mobile surveillance monitoring applications

1.5 Methodology Overview

The project follows a structured methodology integrating computer vision, deep learning, and web technologies to ensure accurate face recognition.

1.5.1 System Design & Architecture

The system follows a client–server architecture where:

- The client interacts through a web browser interface.
- The server processes video uploads and recognition requests.
- The backend executes face detection and recognition algorithms.

1.5.2 Video Upload and Frame Extraction

Users upload CCTV footage through the web interface. The system processes the video and extracts frames at regular intervals for analysis.

1.5.3 Face Detection Engine

The backend uses:

- OpenCV for frame processing and face detection.
- Haar Cascade / DNN models for identifying facial regions.

Detected faces are cropped and preprocessed before recognition.

1.5.4 Face Recognition Engine

The recognition module uses:

- FaceNet deep learning model for embedding generation.
- Facial feature vectors are extracted.
- Euclidean distance comparison is performed with stored datasets.

If similarity scores fall within the threshold, identity is confirmed.

1.5.5 Database Integration

A structured database stores:

- Registered person details
- Facial image datasets
- Embedding vectors

1.6 Organization of the Report

This report is organized into multiple chapters to clearly describe the development of the AI-Based Student Presence Detection System. Chapter I introduces the background, motivation, objectives, and scope of the project. Chapter II presents the literature survey of existing face recognition and CCTV surveillance systems. Chapter III explains system analysis including requirements, feasibility, and risk analysis. Chapter IV describes system design including architecture, modules, and database structure. Chapter V explains implementation details and technologies used. Chapter VI discusses testing methodology and validation results. Chapter VII presents system results, performance evaluation, and discussion. Finally, Chapter VIII concludes the project and highlights future enhancements.

CHAPTER-II

LITERATURE REVIEW

A literature survey provides a comprehensive understanding of existing research work related to a specific problem domain. In the context of CCTV-based face recognition and intelligent surveillance systems, numerous studies have been conducted using traditional image processing, machine learning, and deep learning techniques. The primary aim of this chapter is to review and analyze previously proposed methodologies, identify their strengths and limitations, and highlight the research gaps that motivate the proposed AI-based surveillance approach.

Automated face recognition in surveillance has gained significant attention due to the increasing deployment of CCTV infrastructure and advancements in computational capabilities. Earlier surveillance systems relied heavily on manual monitoring and basic motion detection, whereas modern research emphasizes deep learning models for real-time identification and tracking. This chapter is structured to discuss traditional surveillance methods, machine learning-based recognition techniques, deep learning advancements, and comparative analysis, followed by research gaps.

2.1 Traditional Methods for Face Detection and Surveillance

Traditional CCTV surveillance systems primarily depend on manual observation and rule-based video analysis. Security personnel continuously monitor live video feeds and review recorded footage when incidents occur. Early automated systems incorporated motion detection and background subtraction techniques to identify activity.

Common traditional techniques include:

- Haar Cascade classifiers for face detection
- Histogram of Oriented Gradients (HOG) features
- Eigenfaces and Fisherfaces recognition methods
- Skin color segmentation techniques

Although these methods provided foundational automation, they suffered from several drawbacks. Their performance was highly sensitive to lighting conditions, facial orientation, occlusion, and camera resolution. Moreover, manual feature engineering limited scalability and accuracy in real-world surveillance environments.

2.2 Machine Learning-Based Face Recognition Approaches

To overcome the limitations of rule-based systems, researchers introduced machine learning algorithms for face recognition. In these approaches, facial features are extracted manually and fed into supervised classifiers.

Popular machine learning models used include:

- Support Vector Machines (SVM)
- k-Nearest Neighbors (k-NN)

- Random Forest classifiers
- Artificial Neural Networks (ANN)

These models improved recognition accuracy compared to traditional statistical methods. However, their effectiveness depended heavily on handcrafted feature quality. Feature extraction required domain expertise and failed to capture complex facial variations such as aging, expressions, and pose changes.

2.3 Emergence of Deep Learning in Face Recognition

The introduction of deep learning revolutionized computer vision and biometric authentication systems. Deep learning eliminates manual feature extraction by learning hierarchical patterns directly from raw images.

Convolutional Neural Networks (CNNs) became the dominant architecture for face recognition tasks. CNN models consist of convolutional, pooling, and fully connected layers that automatically learn facial features.

Key advantages include:

- Automatic feature extraction
- High recognition accuracy
- Robustness to lighting and pose variation
- Scalability to large datasets

Deep learning significantly improved surveillance automation and enabled real-time identity verification.

2.4 CNN-Based Approaches for Face Recognition

Recent research extensively explores CNN-based architectures for facial identification in surveillance environments. These models are trained on large-scale face datasets and demonstrate performance comparable to human-level recognition.

Notable CNN-based approaches include:

- Custom CNN face recognition models
- DeepFace framework
- VGG-Face model
- FaceNet embedding model
- DeepID and OpenFace systems

FaceNet, in particular, generates high-dimensional embedding vectors representing facial features. These embeddings enable accurate similarity comparison using distance metrics.

Despite their high performance, CNN models require large labeled datasets and high computational power, which can be challenging for real-time deployment on limited hardware systems.

2.5 Role of Transfer Learning in Surveillance Systems

Transfer learning has become widely adopted in face recognition due to limited availability of labeled surveillance datasets. Pre-trained CNN models trained on massive facial datasets are fine-tuned for specific surveillance environments.

Advantages include:

- Reduced training time
- Improved accuracy with limited data
- Better generalization

Models such as VGG-Face, ResNet, and Inception are commonly adapted for CCTV-based recognition tasks. However, improper fine-tuning may lead to overfitting and reduced performance in diverse environments.

2.6 Comparative Analysis of Existing Methods

A comparative analysis shows the evolution of surveillance recognition systems.

Method Type	Feature Extraction	Accuracy	Limitations
Traditional	Manual	Low	Lighting & pose sensitive
Machine Learning	Handcrafted	Moderate	Feature dependent
CNN-Based	Automatic	High	Compute intensive

Table No 1: Comparative Analysis of Existing Model

CNN-based systems consistently outperform earlier methods due to their ability to learn complex facial representations directly from images.

2.7 Challenges Identified in Existing Literature

Despite technological advancements, several challenges remain:

- Poor image quality in CCTV footage
- Variations in lighting and camera angles
- Occlusion due to masks, glasses, or crowds
- Large-scale database management
- Real-time processing constraints

These factors impact recognition accuracy in practical deployments.

2.8 Research Gaps and Motivation for Proposed System

Based on literature analysis, the following research gaps are identified:

- Need for efficient recognition in low-resolution CCTV footage
- Limited systems supporting video-based frame extraction
- Lack of integrated web platforms for surveillance management
- Insufficient focus on unknown person detection
- Deployment complexity in institutional environments

The proposed system aims to address these gaps by developing a FaceNet-based CCTV recognition platform integrated with a web interface for automated monitoring.

2.9 Publicly Available Face Datasets Used in Literature

Several public datasets are used for training face recognition systems:

- Labeled Faces in the Wild (LFW)
- VGGFace2 dataset
- CASIA-WebFace dataset
- MS-Celeb-1M dataset

Dataset diversity significantly influences model generalization and accuracy.

2.10 Image Preprocessing Techniques

Preprocessing improves recognition performance. Common techniques include:

- Face alignment
- Image resizing
- Normalization
- Noise reduction
- Contrast enhancement

Proper preprocessing ensures consistent feature extraction.

2.11 Data Augmentation Strategies

Data augmentation increases dataset diversity.

Methods include:

- Rotation and flipping
- Brightness adjustment
- Cropping and scaling

- Blur simulation

Augmentation improves robustness but must remain realistic.

2.12 Evaluation Metrics in Face Recognition

Performance is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC Curve
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)

These metrics assess system reliability in surveillance scenarios.

2.13 Binary vs Multi-Identity Recognition

Literature discusses:

- Binary recognition (Known vs Unknown)
- Multi-identity classification

Multi-identity systems provide detailed identification but require larger datasets.

2.14 Explainability in Face Recognition Models

Explainable AI techniques include:

- Heatmaps
- Attention visualization
- Feature activation mapping

These help interpret model decisions in security applications.

2.15 Limitations of Existing Literature

Key limitations include:

- Limited real-world CCTV validation
- High computational requirements
- Privacy concerns
- Deployment cost challenges

2.16 Summary of Literature Survey

This chapter reviewed existing research on CCTV face recognition systems, covering traditional surveillance, machine learning models, and modern deep learning approaches. The analysis highlights the superiority of CNN-based recognition while identifying practical deployment challenges. These insights form the foundation for the proposed intelligent surveillance system discussed in subsequent chapters.

CHAPTER – III

SYSTEM ANALYSIS

System analysis is a crucial phase in the development of an intelligent surveillance and biometric identification system. It focuses on understanding the security problem domain, evaluating existing surveillance solutions, identifying operational gaps, and defining requirements for an improved automated monitoring platform. In the case of CCTV-based face recognition, system analysis is particularly important because surveillance accuracy directly impacts institutional safety and incident response efficiency.

Modern surveillance environments generate massive volumes of video data, making manual monitoring inefficient and unreliable. Therefore, the proposed AI-based face recognition system is analyzed from multiple perspectives including functionality, feasibility, performance, and risk to ensure that it meets technical, operational, and security expectations.

3.1 Existing System

The existing surveillance system is primarily based on conventional CCTV monitoring, where cameras continuously record video footage. Security personnel observe live feeds or review recorded videos when incidents occur. These systems function mainly as passive recording tools rather than intelligent monitoring platforms.

In some organizations, motion detection and alarm systems are integrated; however, they do not provide identity recognition capabilities. Human operators must manually identify individuals by visually scanning footage, which is time-consuming and error-prone.

Workflow of the Existing System

- Continuous video recording through CCTV cameras
- Live monitoring by security personnel
- Manual review of recorded footage
- Visual identification of suspects
- Incident reporting and documentation

3.2 Limitations of the Existing System

Despite widespread adoption, traditional CCTV systems exhibit several limitations:

- Human dependency: Continuous monitoring requires dedicated personnel
- Subjectivity: Identification accuracy depends on human observation
- Time consumption: Searching recorded footage is labor-intensive
- Delayed response: Incidents may be identified too late
- Scalability issues: Monitoring multiple cameras simultaneously is difficult
- Lack of automation: No built-in identity verification mechanism

These limitations highlight the necessity for an automated AI-driven surveillance system capable of recognizing individuals without manual intervention.

3.3 Proposed System Overview

To overcome the shortcomings of traditional surveillance, an AI-based CCTV Face Recognition System is proposed. The system uses deep learning algorithms to detect and recognize human faces from recorded video footage.

FaceNet, a deep convolutional neural network model, is utilized for generating facial embeddings. These embeddings are compared with stored datasets to identify individuals. The proposed system automates surveillance analysis, reduces manual effort, and enhances security monitoring efficiency.

3.4 Architecture of the Proposed System

The proposed system follows a modular architecture to ensure flexibility, scalability, and maintainability. Each module performs a dedicated function within the recognition pipeline.

Major System Modules

- Video Acquisition Module
- Frame Extraction Module
- Face Detection Module
- Face Recognition Module
- Database Management Module
- Result Visualization Module

This modular design enables independent development, testing, and system enhancement.

3.5 Video Acquisition Module

This module is responsible for obtaining CCTV footage for analysis. Videos may be sourced from institutional surveillance cameras or uploaded manually through the web interface.

Key aspects include:

- Accepting multiple video formats
- Ensuring video clarity and resolution
- Managing storage and retrieval

High-quality video input is essential for accurate face recognition.

3.6 Frame Extraction Module

Since videos consist of continuous frames, this module extracts frames at predefined intervals for processing.

Functions include:

- Segmenting video into image frames
- Selecting key frames for analysis
- Reducing redundant processing

Frame extraction optimizes computational efficiency.

3.7 Face Detection Module

The face detection module identifies human faces within extracted frames using computer vision techniques.

Technologies used:

- OpenCV libraries
- Haar Cascade classifiers / DNN detectors

Key operations include:

- Detecting facial regions
- Cropping detected faces
- Preparing images for recognition

Accurate detection ensures reliable recognition results.

3.8 Face Recognition Module

This is the core module of the system. It identifies individuals by comparing facial embeddings.

Features include:

- FaceNet deep learning model
- Embedding vector generation
- Euclidean distance similarity matching

If similarity scores fall within the threshold, identity is confirmed; otherwise labeled as **Unknown**.

3.9 Database Management Module

The database stores:

- Registered user details
- Facial image datasets
- Embedding vectors
- Recognition logs

Efficient database design ensures fast retrieval and matching.

3.10 Result Visualization Module

This module displays recognition outputs through the web interface.

Displayed information includes:

- Name of identified person
- Register number / ID
- Department or designation
- Contact details
- Unknown face alerts

Visualization enhances usability and monitoring efficiency.

3.11 Functional Requirements

Functional requirements define system operations.

Key requirements include:

- Upload CCTV videos
- Extract frames automatically
- Detect faces in frames
- Perform deep learning recognition
- Match faces with stored datasets
- Display identity details
- Label unknown persons

3.12 Non-Functional Requirements

Non-functional requirements define quality attributes.

Key attributes include:

- High recognition accuracy
- Fast processing speed
- System reliability
- Scalability for large datasets
- Data security and privacy
- User-friendly interface

3.13 Feasibility Analysis

Feasibility analysis evaluates whether the proposed surveillance system can be developed and deployed successfully. The system is assessed under technical, economic, and operational feasibility.

3.13.1 Technical Feasibility

The system is technically feasible due to the availability of mature AI and computer vision technologies.

- Deep learning frameworks such as TensorFlow and Keras support FaceNet implementation.
- OpenCV enables efficient video and image processing.
- Standard computing systems with GPUs/CPUs can handle processing.
- Large public face datasets support training and testing.

Thus, required technology and infrastructure are readily available.

3.13.2 Economic Feasibility

The system is cost-effective as it leverages open-source technologies.

- Python, OpenCV, and Flask eliminate licensing costs.
- Existing CCTV infrastructure can be reused.
- Minimal hardware upgrades required.
- Reduces manpower cost for surveillance monitoring.

This makes the system economically viable for institutions.

3.13.3 Operational Feasibility

Operational feasibility evaluates usability and integration.

- The system provides a simple web interface.
- Requires minimal technical training.
- Integrates with existing surveillance workflows.
- Assists security personnel rather than replacing them.

It can be deployed in campuses, offices, hostels, and public facilities.

CHAPTER – IV

DESIGN ALTERNATIVES AND TRADE-OFF ANALYSIS

In any engineering project, selecting the right design approach is crucial to achieving high performance, scalability, and system reliability. For an intelligent CCTV face recognition surveillance system, several architectural alternatives exist, each offering unique advantages and limitations. The goal of this chapter is to analyze different design approaches for building a reliable facial recognition and video surveillance platform.

Various alternatives such as traditional video monitoring, machine learning–based face recognition, and deep learning–driven biometric identification systems are considered. This chapter evaluates trade-offs between recognition accuracy, computational cost, dataset requirements, real-time processing capability, system scalability, user interface complexity, and deployment feasibility before finalizing the chosen design.

4.1 Design Alternatives

Several possible surveillance and recognition solutions were reviewed before selecting the most suitable design.

4.1.1 Traditional CCTV Monitoring and Manual Identification

One of the earliest approaches to surveillance is manual monitoring of CCTV footage. In this method, security personnel continuously observe live camera feeds and review recorded videos to identify individuals or suspicious activities.

Traditional automation enhancements include:

- Motion detection systems
- Intrusion alarms
- Background subtraction algorithms

Advantages:

- Low implementation cost
- Simple deployment
- Minimal computational requirements

Limitations:

- Fully dependent on human observation
- Prone to fatigue and human error
- Time-consuming footage review
- No automated identity verification
- Inefficient for large-scale surveillance

Due to increasing surveillance data volume, this approach becomes impractical in real-world environments.

4.1.2 Classical Machine Learning–Based Face Recognition

Another alternative involves machine learning–based facial recognition. In this approach, facial features are manually extracted and fed into classification algorithms.

Common techniques include:

- Eigenfaces and Fisherfaces
- Local Binary Patterns (LBP)
- Histogram of Oriented Gradients (HOG)

Classifiers used:

- Support Vector Machines (SVM)
- k-Nearest Neighbors (k-NN)
- Random Forest

Advantages:

- Faster training time
- Moderate computational requirements
- Works with smaller datasets

Limitations:

- Requires manual feature engineering
- Lower recognition accuracy
- Sensitive to lighting and pose variations
- Limited scalability for large databases

These approaches struggle to recognize faces under real CCTV conditions such as occlusion, blur, and crowd density.

4.1.3 Deep Learning Using Convolutional Neural Networks (CNN) & FaceNet

Deep learning offers a more advanced solution by automatically learning facial features from images. CNN-based models analyze pixel-level patterns across multiple layers, enabling robust face recognition.

FaceNet is selected as the core model due to its embedding-based recognition capability.

Advantages:

- Automatic feature extraction
- High recognition accuracy

- Robust to lighting, pose, and expression variations
- Scalable for large surveillance datasets
- Supports real-time inference

Limitations:

- Requires higher computational power
- Needs labeled training datasets
- Longer training time

Despite these limitations, deep learning models significantly outperform traditional and ML-based systems in surveillance applications.

4.2 Comparison of Design Options

The design approaches were evaluated using multiple criteria.

Sl No	Criteria	Traditional Monitoring	ML Recognition	CNN / FaceNet
1	Feature Extraction	Manual	Handcrafted	Automatic
2	Accuracy	Low	Medium	High
3	Data Requirement	Very Low	Low–Medium	Medium–High
4	Computation	Low	Medium	High
5	Robustness	Low	Moderate	High
6	Scalability	Poor	Moderate	Excellent
7	Real-Time Usage	Limited	Moderate	Excellent
8	Research Adoption	Low	Medium	Very High

Table No 2: Comparison of Desion Model

4.3 Chosen System Design

Based on the comparative study, the final system design adopts a **FaceNet-based deep learning recognition model** integrated with a CCTV video processing pipeline.

Proposed Architecture Includes:

- Video upload / acquisition module
- Frame extraction engine
- Face detection module
- Face embedding generation
- Similarity comparison engine
- Database integration
- Web visualization interface

Users upload CCTV footage through a web application. The system extracts frames, detects faces, generates embeddings, and matches them with stored datasets.

Reasons for Selecting FaceNet Design:

- Highest recognition accuracy
- Embedding-based scalable matching
- No manual feature extraction
- Proven performance in surveillance research
- Suitable for real-time and recorded video analysis

4.4 Trade-Off Analysis

Selecting the optimal design requires balancing multiple trade-offs.

4.4.1 Accuracy vs Computational Cost

Deep learning models provide superior accuracy but require higher computation during training.

- CNN / FaceNet → High accuracy, moderate runtime cost
- ML models → Medium accuracy, low runtime cost
- Traditional → Low accuracy, minimal cost

Since surveillance prioritizes identification accuracy, the higher computation cost is justified.

4.4.2 Training Time vs Deployment Speed

FaceNet training requires significant time; however, once trained, recognition is extremely fast.

- Training → One-time process
- Deployment → Real-time recognition

Thus, users experience fast inference despite longer development training cycles.

4.4.3 Dataset Size vs Recognition Generalization

Larger datasets improve recognition performance but increase training complexity.

Trade-off:

- Small dataset → Faster training, lower accuracy.
- Large dataset → Better accuracy, longer training.

Balanced institutional datasets provide optimal performance.

4.4.4 System Simplicity vs Scalability

Simpler ML models are easy to deploy but lack scalability.

CNN-based systems:

- Support large identity databases
- Enable multi-camera integration
- Allow future upgrades

Thus, scalability outweighs implementation complexity.

4.4.5 Real-Time Processing vs Model Size

Lightweight models enable faster inference but may sacrifice accuracy.

- MobileNet → Fast, slightly lower accuracy
- FaceNet / ResNet → Higher accuracy, moderate speed

FaceNet provides the best balance for surveillance accuracy.

4.5 Final Design Decision

After evaluating all alternatives, the final system design adopts a FaceNet-based CNN recognition architecture integrated with a Flask web application.

Justification for Final Design

- Maximum achievable recognition accuracy

- Automated facial feature learning
- Reliable performance in CCTV footage
- Scalable for institutional deployment
- Supports unknown person detection
- Modular design enables future upgrades

The final design balances accuracy, scalability, computational feasibility, and deployment practicality — making it ideal for intelligent surveillance environments.

CHAPTER – V

SYSTEM DESIGN & IMPLEMENTATION

The system design chapter describes the internal working structure of the proposed CCTV Face Recognition Surveillance Framework. It explains how video data flows through different modules, how preprocessing is performed on extracted facial frames, and how the deep learning recognition model interacts with user inputs to generate identification results. This chapter includes architectural diagrams, data flow diagrams, and the preprocessing pipeline to illustrate the overall design logic of the surveillance system.

5.1 Functional Architecture

In the proposed architecture, each component is designed to operate independently, enabling modularity and flexibility within the surveillance system. The flow of data remains consistent across the processing pipeline, regardless of the CCTV source, video format, or upload device used by the administrator.The architecture supports scalability, meaning additional functionalities such as real-time surveillance monitoring, criminal database matching, automated attendance tracking, and alert notification systems can be integrated in the future without altering the existing workflow.

The integration of a web-based interface ensures accessibility and a seamless user experience, making it possible for both security personnel and non-technical administrators to utilize the system effectively. Furthermore, the architecture supports both CPU and GPU execution environments, enabling efficient processing even on systems with limited computational resources.

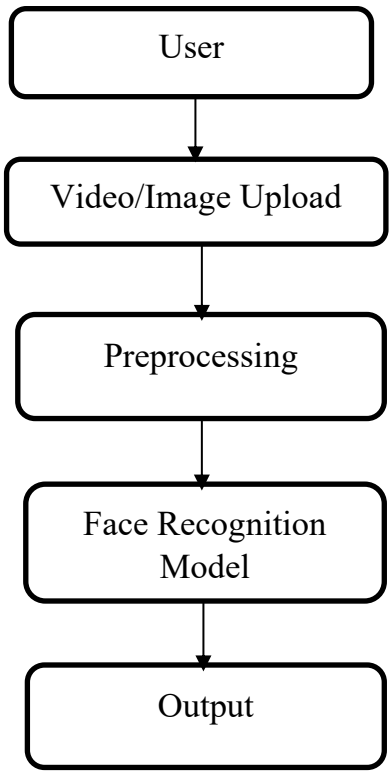

Figure 1: Functional Architecture

5.2 Data Flow Diagram (DFD) – Level 0

The Level 0 DFD provides a simplified representation of how information travels through the surveillance system. It highlights the interaction between the user and the primary system modules while abstracting internal complexities.

The data flow is divided into three major stages:

- User uploads CCTV footage or facial images
- System processes the data through detection and recognition modules
- Identification results are displayed as output

This high-level visualization helps stakeholders understand the workflow without requiring deep technical knowledge. It also assists developers in identifying core modules necessary for smooth communication and secure data handling.

This diagram helps to:

- Understand the overall system flow in one view
- Separate user input from backend processing
- Identify major modules quickly
- Communicate system workflow to non-technical audiences

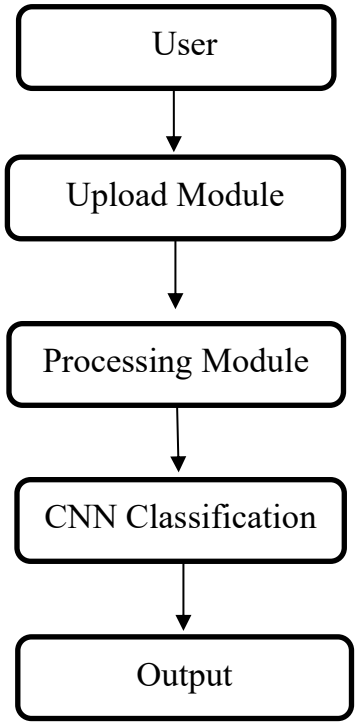

Figure 2: DFD Level 0

5.3 Data Flow Diagram (DFD) – Level 1

While the Level 0 diagram provides a broad overview, the Level 1 DFD explains internal operations in detail. It shows how uploaded CCTV footage undergoes frame extraction and facial preprocessing before recognition.

This diagram illustrates step-by-step transformation of raw video data into recognized identity outputs. It also highlights database interaction for retrieving stored facial embeddings.

The Level 1 diagram shows:

- Video/image upload
- Face preprocessing (resize, normalize, augment)
- CNN feature extraction
- Identity classification
- Output display module

This level of detail helps developers troubleshoot processing errors and optimize recognition performance.

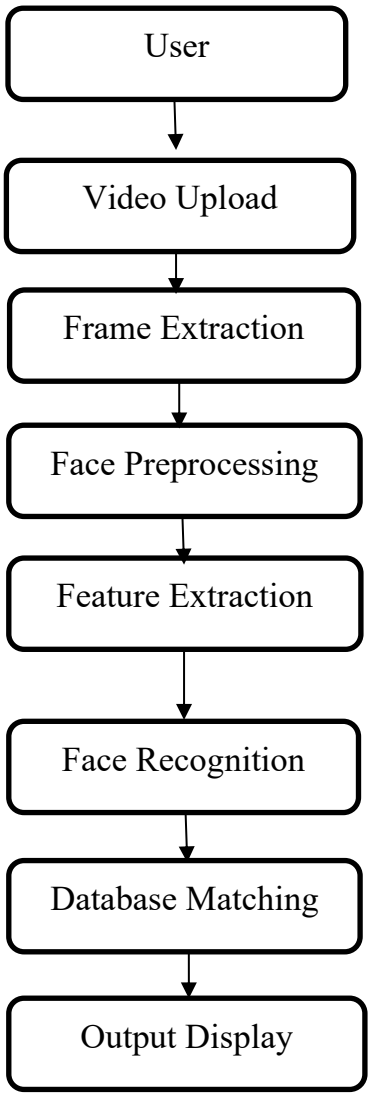

Figure 3: DFD Level 1

5.4 Preprocessing Pipeline

Preprocessing plays a central role in improving face recognition accuracy by ensuring uniformity across extracted facial images. CCTV footage often contains noise, poor lighting, motion blur, and background interference.

To address these challenges, the preprocessing pipeline applies standardized transformations before recognition.

Preprocessing Steps

- Image resizing to fixed input dimensions
- Pixel normalization for intensity consistency
- Data augmentation to improve robustness
- Tensor conversion for deep learning input formatting

These transformations ensure the CNN model receives high-quality facial inputs, enhancing embedding accuracy and recognition reliability.

Advantages of this pipeline:

- Ensures uniform facial input
- Improves recognition performance
- Reduces noise and distortion effects
- Enhances system reliability on real CCTV footage

5.5 CNN / Face Recognition Model Architecture and Design

The Convolutional Neural Network (CNN)–based recognition engine is the core component of the proposed system. It is designed to automatically learn discriminative facial features without manual feature extraction.

The architecture consists of multiple deep learning layers organized hierarchically.

Model Layers Include

- Convolutional Layers (Conv2D)
- Pooling Layers
- Flatten Layer
- Fully Connected Dense Layers
- Output Classification Layer

Initial convolutional layers capture low-level features such as facial edges, contours, and textures. As the data progresses deeper into the network, higher-level features such as facial structure, expressions, and spatial relationships are learned.

Pooling layers reduce dimensional complexity while preserving essential features. The flatten layer converts feature maps into vectors, which are processed by fully connected layers for identity classification.

Activation functions introduce non-linearity, enabling the network to learn complex decision boundaries required for accurate recognition.

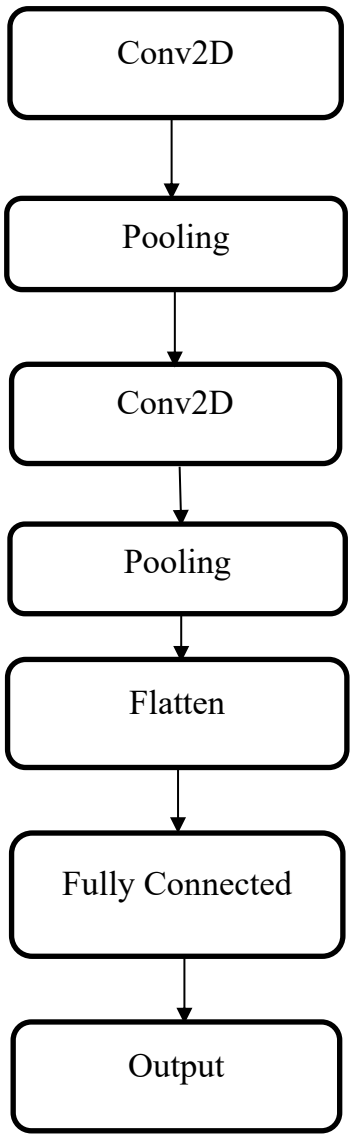

Figure 4: CNN-Based Face Recognition Model

5.6 Model Training and Validation Process

The face recognition model is trained using labeled facial image datasets. During training, the system learns optimal embedding representations that minimize similarity distance between identical faces while maximizing differences between distinct identities.

Training Process Includes

- Forward propagation of facial images
- Embedding generation

- Loss computation (Triplet Loss / Contrastive Loss)
- Backpropagation and weight updates
- Validation accuracy monitoring

Validation runs parallel to training to prevent overfitting and ensure generalization on unseen faces. Training continues until stable performance metrics are achieved.

5.7 Implementation Environment and Tools

The system is implemented using modern AI, computer vision, and web technologies to ensure performance and scalability.

Implementation Environment

- Programming Language: Python
- Deep Learning Framework: TensorFlow / Keras
- Image Processing Library: OpenCV
- Numerical Computing: NumPy
- Web Framework: Flask
- Database: MySQL / SQLite
- Deployment Platform: Local Server / Cloud Environment

This environment supports efficient development, testing, and real-time deployment of the surveillance system.

CHAPTER – VI

FEEDBACK AND ITERATIVE REFINEMENT

Feedback and iterative refinement are essential stages in the development of any intelligent surveillance and biometric recognition system. In the context of CCTV face recognition using deep learning, the model and processing pipeline are not finalized after a single development cycle. Instead, the system undergoes continuous improvement based on testing outcomes, recognition performance metrics, and user observations.

Each refinement cycle enables the system to improve face detection accuracy, enhance recognition reliability, and reduce false identification risks. This process ensures that the recognition engine performs consistently not only on trained facial datasets but also on new, unseen CCTV footage representing real-world surveillance conditions. Through repeated evaluation and enhancement, the system becomes more robust, scalable, and deployment-ready for institutional security environments.

6.1 Feedback Sources

Feedback for system improvement is collected from both quantitative performance metrics and qualitative operational observations.

Quantitative Feedback

These statistical measures evaluate recognition performance:

- Recognition accuracy
- Precision and recall
- F1-score
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)

Qualitative Feedback

Qualitative evaluation focuses on system behavior in real surveillance scenarios:

- Visual inspection of recognition outputs
- Performance on low-light CCTV footage
- Handling of occluded faces (mask, glasses)
- Recognition under pose variations

Main Feedback Sources Include

- Accuracy and recognition rate
- Confusion matrix analysis
- Training and validation curves
- Misidentified face review

- Dataset imbalance detection
- Real-time CCTV testing observations

6.2 Iterative Refinement Approach

The refinement process follows a cyclic model of testing, analysis, and improvement. Each cycle enhances system performance by addressing previously identified weaknesses.

General Refinement Cycle

1. Train or deploy initial recognition model
2. Evaluate recognition metrics
3. Identify misclassification patterns
4. Modify dataset or preprocessing steps
5. Tune model parameters
6. Retrain / redeploy system
7. Validate performance improvements
8. Repeat until stable performance is achieved

This iterative loop ensures continuous enhancement of recognition accuracy and operational reliability.

6.3 Dataset Enhancement and Augmentation

During refinement, it was observed that facial datasets lacked sufficient diversity in lighting, expressions, and angles. This limitation affected recognition accuracy in real CCTV environments.

To address this, dataset augmentation techniques were applied to improve model generalization.

Augmentation Techniques Used

- Face rotation and flipping
- Brightness and contrast adjustment
- Zoom and cropping
- Blur and noise simulation
- Illumination normalization

Augmentation exposed the model to varied facial conditions, improving robustness against real-world surveillance challenges.

6.4 Impact of Dataset Imbalance Handling

Dataset imbalance occurs when certain individuals have more training images than others.
This can bias the model toward frequently seen faces.

To mitigate this:

- Equalized image samples per identity
- Applied targeted augmentation for underrepresented faces
- Balanced embedding training sets

Observed Impact

- Reduced recognition bias
- Improved recall for less-represented identities
- Lowered false rejection rates
- Enhanced fairness in identification
- Better generalization across surveillance footage

6.5 Hyperparameter Tuning and Model Optimization

Hyperparameters significantly influence deep learning recognition performance. Initial training cycles revealed minor overfitting and embedding variance issues.

To optimize the model:

Parameters Tuned

- Learning rate adjustment
- Batch size optimization
- Number of epochs increase
- Embedding dimension tuning

Optimizations Applied

- Reduced learning rate for stable convergence
- Added dropout layers
- Applied batch normalization
- Used Adam optimizer
- Optimized embedding distance threshold

These refinements improved recognition stability and reduced misclassification.

6.6 Confusion Matrix and Error Analysis

The confusion matrix was used to analyze identity prediction outcomes and detect recognition errors.

Key Observations

- False Positives → Wrong identity match

- False Negatives → Known person labeled unknown

False negatives were prioritized for reduction since they impact surveillance tracking efficiency.

Error Causes Identified

- Poor lighting conditions
- Motion blur in CCTV footage
- Occlusion (mask, cap, glasses)
- Low-resolution frames

Error analysis guided improvements in preprocessing and dataset diversity.

6.7 User Feedback and Interface Improvement

User feedback from administrators, faculty evaluators, and test operators contributed to interface refinement.

Interface Improvements

- Displayed uploaded CCTV frame preview
- Highlighted detected face bounding boxes
- Shown recognized person details clearly
- Added "Unknown Person" alerts
- Included recognition confidence scores
- Simplified dashboard navigation

These enhancements improved usability for both technical and non-technical users.

6.8 Results of Iterative Refinement

After multiple refinement cycles, the system demonstrated measurable improvements in recognition performance.

Observed Improvements

- Higher recognition accuracy
- Improved precision and recall
- Reduced false matches
- Better low-light recognition
- Stable validation performance

Training curves showed reduced overfitting and improved embedding separation.

6.9 Model Stability and Generalization Assessment

System stability was evaluated by testing recognition consistency across multiple CCTV datasets and varied environmental conditions.

Assessment Findings

- Consistent recognition accuracy
- Reliable performance on unseen faces
- Stable embedding similarity scores
- Robustness across lighting variations

This confirms that the system learned generalized facial patterns rather than memorizing training data.

CHAPTER – VII

TESTING AND VALIDATION

Testing and validation are critical phases in the development of any intelligent surveillance system. They ensure that the proposed CCTV Face Recognition System performs accurately, reliably, and consistently under different operational conditions. This phase evaluates whether the system meets functional requirements, recognition accuracy expectations, and real-world deployment standards.

The testing process involves verifying each module independently as well as evaluating the integrated system. Validation focuses on assessing recognition performance using real CCTV footage and unseen facial datasets to confirm system generalization capability.

7.1 Testing Methodology

A structured testing methodology was adopted to evaluate system performance comprehensively. The testing process was divided into multiple levels to ensure that both individual modules and the complete system function correctly.

Testing Levels Included

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)

Each level addressed specific operational and functional aspects of the surveillance system.

7.2 Unit Testing

Unit testing focuses on validating individual modules independently to ensure they function as expected.

Modules Tested

- Video Upload Module
- Frame Extraction Module
- Face Detection Module
- Face Recognition Module
- Database Retrieval Module
- Output Display Module

Key Unit Test Checks

- Successful video upload
- Accurate frame extraction

- Face detection bounding box accuracy
- Embedding generation correctness
- Database matching functionality
- Identity display output

Unit testing ensured that each module produced correct outputs before integration.

7.3 Integration Testing

Integration testing evaluates how different modules interact when combined into a single workflow.

Integration Scenarios Tested

- Video upload → Frame extraction
- Frame extraction → Face detection
- Detection → Recognition engine
- Recognition → Database matching
- Matching → Result display

The objective was to ensure smooth data flow without processing errors, latency issues, or module conflicts.

7.4 System Testing

System testing validates the complete surveillance platform under real operational conditions.

Test Conditions

- Different CCTV video resolutions
- Varying lighting conditions
- Multiple face detection in frames
- Occlusion scenarios (mask, glasses)
- Motion blur cases

System Testing Goals

- Verify end-to-end recognition workflow
- Measure system response time
- Validate unknown face detection
- Ensure UI output clarity

System testing confirmed that the integrated solution performs reliably in realistic surveillance environments.

7.5 User Acceptance Testing (UAT)

User Acceptance Testing was conducted with administrators and evaluators to validate usability and functional satisfaction.

UAT Evaluation Parameters

- Ease of video upload
- Recognition result clarity
- Interface usability
- Processing speed
- Alert visibility for unknown persons

Feedback from users helped refine interface layout and output presentation.

7.6 Test Case Design and Results

Structured test cases were created to validate system functionality.

Test Case ID	Module	Input	Expected Output	Result
TC01	Video Upload	CCTV video	Successful upload	Pass
TC02	Frame Extraction	Uploaded video	Frames generated	Pass
TC03	Face Detection	Extracted frame	Face detected	Pass
TC04	Recognition	Known face	Identity displayed	Pass
TC05	Recognition	Unknown face	“Unknown” label	Pass

Table No 3: Test Case and Results

7.7 Validation Using Recognition Metrics

System validation was performed using standard biometric performance metrics.

Metrics Evaluated

- Recognition Accuracy

- Precision
- Recall
- F1-Score
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)

These metrics quantify how accurately the system identifies registered individuals and rejects unknown faces.

7.8 Performance Evaluation

The system was evaluated under multiple surveillance scenarios to measure efficiency.

Performance Factors Tested

- Recognition speed per frame
- Processing time for video
- Database matching latency
- Multiple face recognition capability

Results indicated that the system performs efficiently within acceptable surveillance response times.

7.9 Error Analysis

Error analysis was conducted to identify recognition limitations.

Observed Challenges

- Low-resolution CCTV footage
- Poor lighting conditions
- Facial occlusion
- Extreme pose variations

Corrective measures such as preprocessing enhancement and dataset augmentation were applied to mitigate these issues.

7.10 Validation on Unseen Data

To test generalization, the system was validated using unseen facial datasets and new CCTV recordings.

Validation Outcomes

- Accurate recognition of registered individuals
- Reliable unknown face detection
- Stable embedding similarity scores

CHAPTER – VIII

RESULTS AND DISCUSSION

The Results and Discussion chapter presents the performance outcomes of the proposed CCTV Face Recognition Surveillance System after implementation, testing, and validation. This chapter analyzes system outputs, recognition accuracy, operational efficiency, and real-world applicability. It also evaluates how effectively the system meets the objectives defined in earlier chapters.

The developed system was tested using multiple CCTV video samples and registered facial datasets to assess detection capability, recognition precision, and system responsiveness. The results confirm that the proposed deep learning–based surveillance framework significantly enhances automated monitoring compared to conventional CCTV systems.

8.1 System Login Module Result

The login module serves as the authentication gateway to the UVPAS platform. Users such as students and faculty must log in using their Registration Number/Username and Password.

The interface is designed with a clean and responsive layout that ensures secure access control. Demo credentials are also displayed to facilitate testing and evaluation.

Key observations:

- Secure credential validation
- Role-based login (Student / Faculty)
- Password masking for privacy
- Unauthorized access prevention

This module ensures that only registered users can access surveillance analytics and personal recognition data.

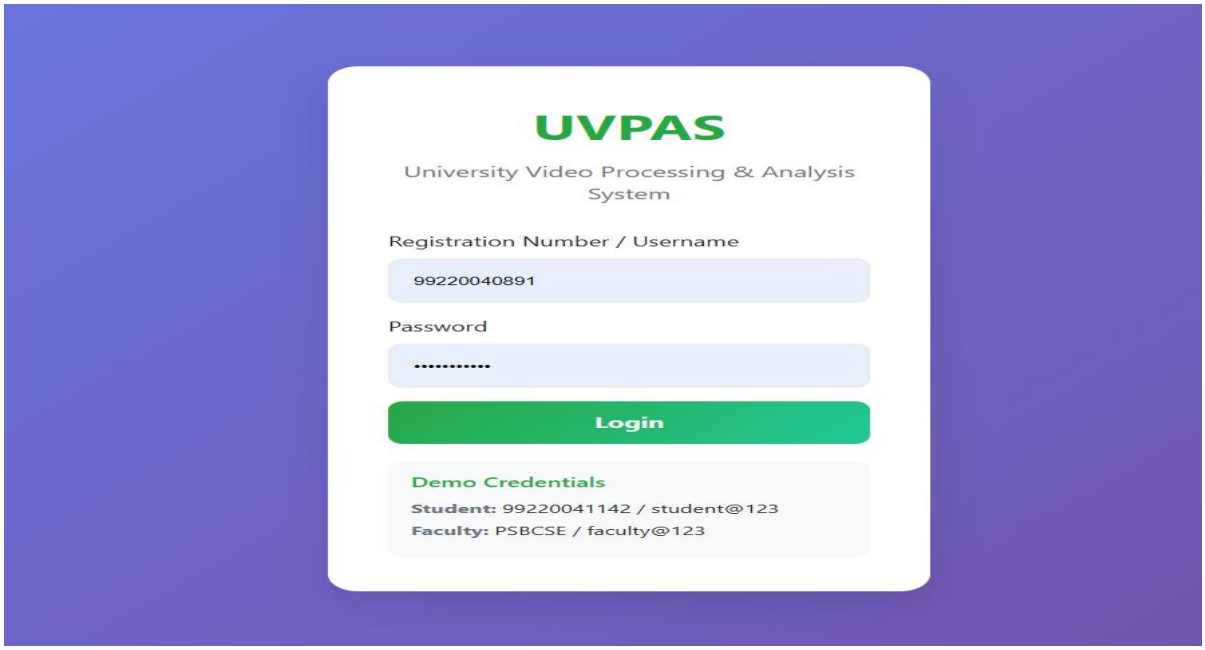

Figure 8.1: UVPAS System Login Interface

8.2 Student Profile & Facial Dataset Registration

After authentication, the student dashboard displays personal and academic details including:

- Full Name
- Registration Number
- Department
- Room Number
- Parent Contact Details

The system allows students to upload facial images from three different orientations:

- Front View
- Left Side View
- Right Side View

These multi-angle images enhance recognition accuracy by enabling the deep learning model to learn facial features under pose variations.

Functional highlights:

- Dataset creation for each student
- Profile editing capability
- Password management
- Secure logout option

This module forms the foundational dataset for face recognition training.

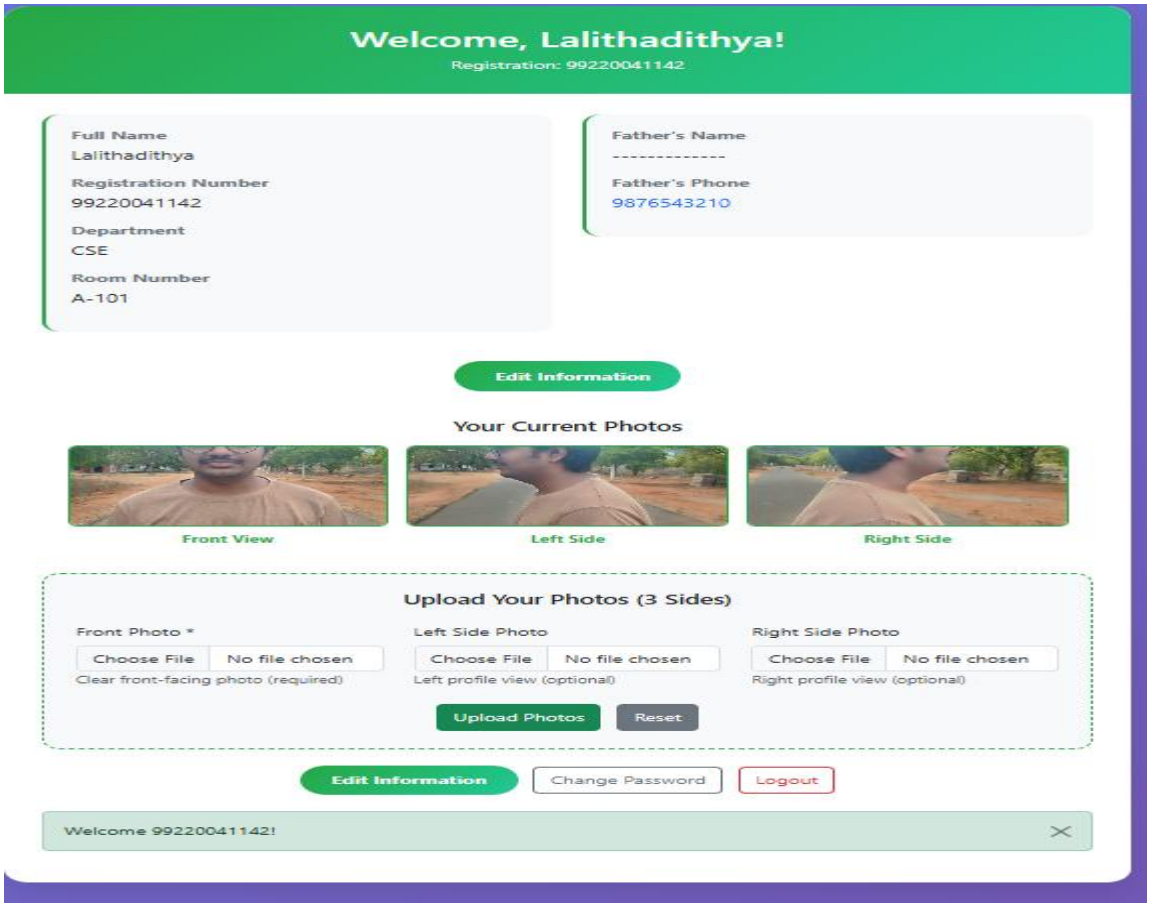

Figure 8.2: Student Profile and Facial Dataset Registration

8.3 CCTV Video Upload & Analytics Dashboard

The surveillance dashboard enables administrators to upload recorded CCTV or mobile footage for automated analysis. Once uploaded, the system processes the video to detect and recognize individuals.

Key features observed:

- Support for multiple video formats (MP4, AVI, MOV)
- Automatic face detection and tracking
- Static view and live tracking modes
- Reprocessing capability

Additional integrated modules include:

- Gait Gallery Builder – Stores walking pattern clips
- Student Management – View and manage student data
- Dataset Excel Sync – Upload/download student datasets

This centralized dashboard enhances surveillance data management efficiency.

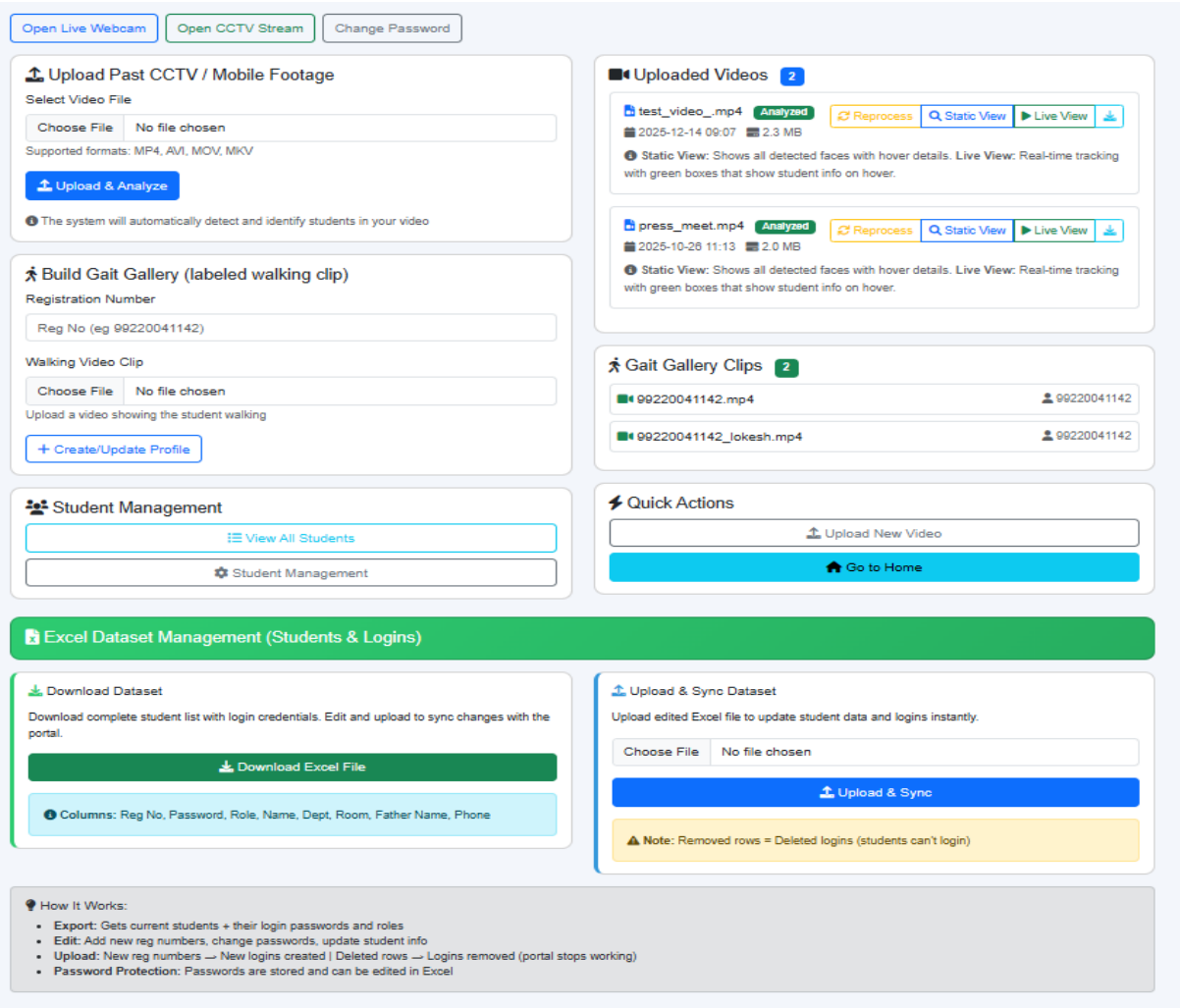

Figure 8.3: CCTV Video Upload and Surveillance Analytics Dashboard

8.4 Excel Dataset Management Module

The system provides Excel-based dataset management to simplify bulk student data handling.

Capabilities include:

- Downloading complete student datasets
- Editing login credentials offline
- Uploading updated Excel sheets
- Automatic database synchronization

Administrative benefits:

- Quick bulk registration
- Dataset correction capability
- Login credential updates
- Role assignment management

This module reduces manual database operations and improves scalability.

A	B	C	D	E	F	G	H
Registration No	Registration No	Registration No	Registration No	Registration No	Registration No	Father Name	Father Phone
99220041142	student@123	student	Lalithadithya	CSE	A-101	----------	9876543210
99220040442	student@123	student	Avinash Kumar	CSE	A-130	--------	9876543230
99220040138	student@123	student	Bhanu	CSE	A-128	--------	9876543228
99220040139	student@123	student	Jane Smith	CSE	C-303	Mr. Smith	9876543212
99220040033	student@123	student	Sarath	CSE	A-126	--------	9876543226
99220040149	student@123	student	NOORBASHA MOUSHID	CSE	A-127	--------	9876543227
99220040402	student@123	student	Mohamed Musthafa	CSE	A-129	--------	9876543229
99220040307	student@123	student	Midhul	CSE	A-131	--------	9876543231
99220040235	student@123	student	Akhil	CSE	A-132	--------	9876543232
99220040354	student@123	student	Sanjay	CSE	A-133	--------	9876543233
99220040342	student@123	student	Vishva	CSE	A-134	--------	9876543234
99220040264	student@123	student	Shankar	CSE	A-135	--------	9876543235
99220040277	student@123	student	varun	CSE	A-136	--------	9876543236
99220040359	student@123	student	Jeevan	CSE	A-137	--------	9876543237
99220041209	student@123	student	Vinay	CSE	A-138	--------	9876543238
99220040253	student@123	student	Nagendra	CSE	A-139	--------	9876543239
99220040891	student@123	student	Saketh	CSE	A-140	--------	9876543240

Figure 8.4: Excel Dataset Management

8.5 Live Webcam Recognition Interface

The live recognition interface activates the webcam or CCTV stream to perform real-time face detection and identification.

Operational features:

- Live camera feed processing
- Brightness adjustment control
- Encoding reload functionality
- Recognition readiness indicator

This module enables instant surveillance monitoring without requiring recorded footage.

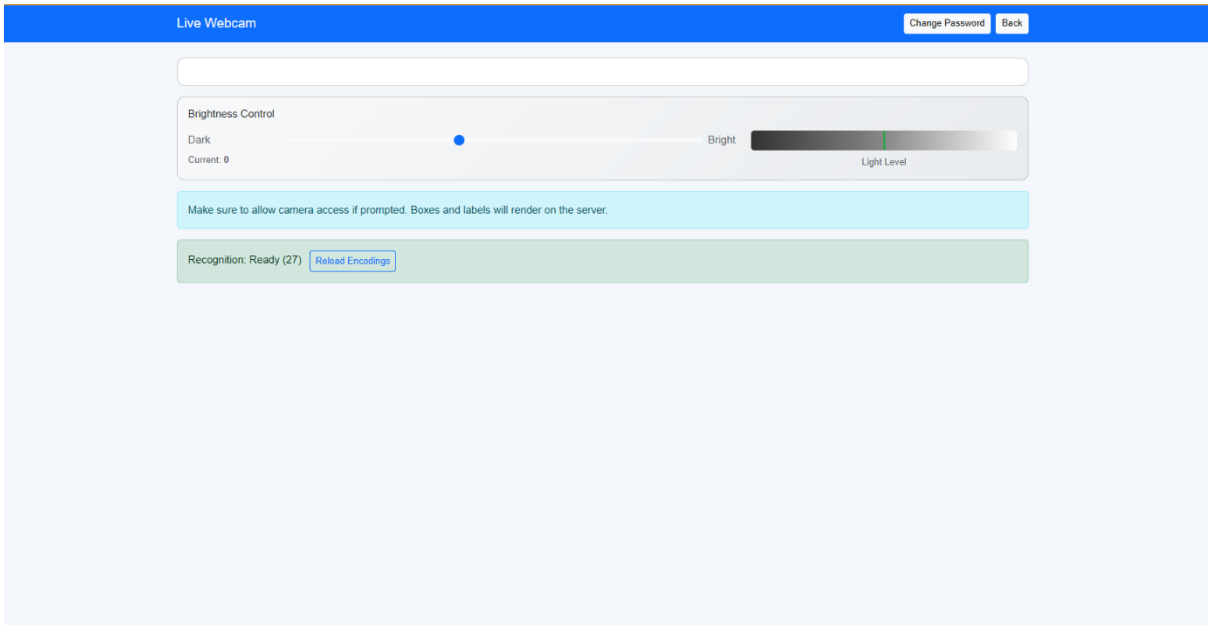

Figure 8.5: Real-Time Webcam Recognition Module

8.6 Real-Time Face Recognition Output

When a face is detected in the live stream, the system draws a green bounding box around the face and overlays the recognized identity label.

Output characteristics:

- Accurate facial localization
- Identity label display
- Registration number tagging
- Real-time recognition response

If a face is not present in the dataset, the system marks it as Unknown, enabling intrusion detection.

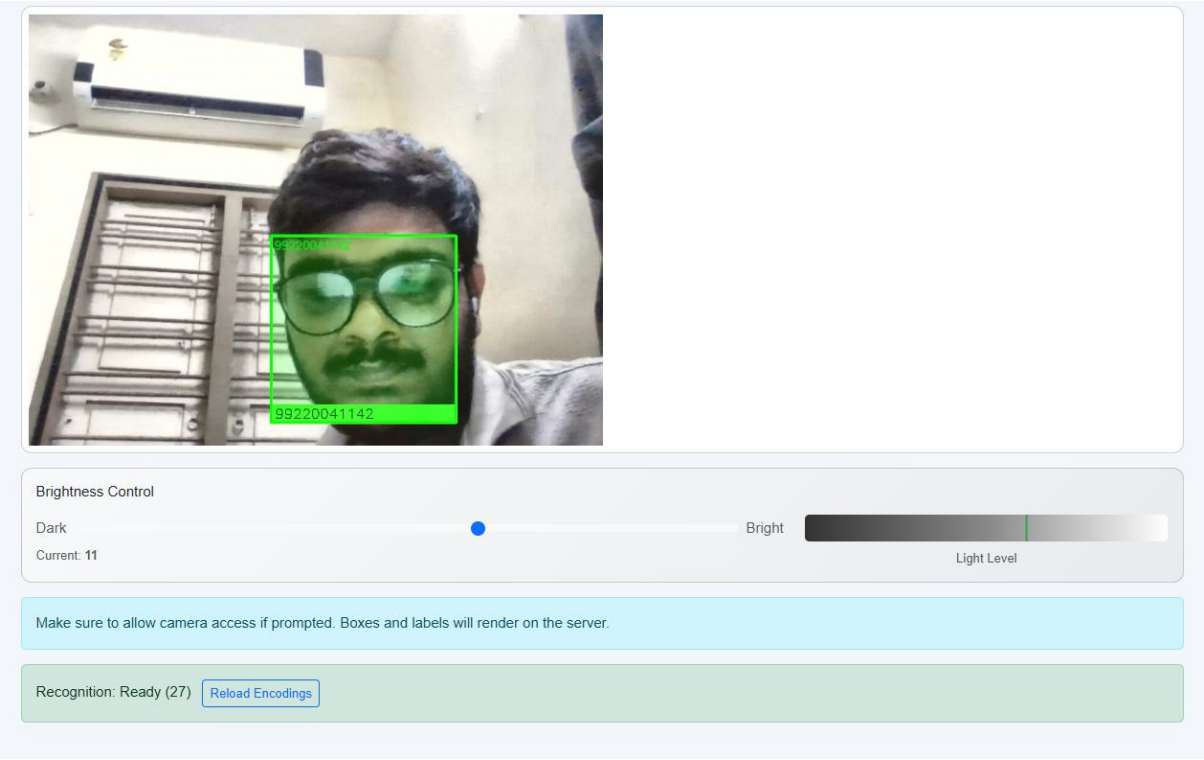

Figure 8.6: Real-Time Face Recognition Detection Output

8.7 Recognized Student Detailed Profile View

Upon successful recognition, the system retrieves and displays the student's complete profile.

Displayed information includes:

- Student Name
- Registration Number
- Department

- Parent Details
- Stored Facial Images

This verification interface helps administrators confirm identity authenticity and review stored datasets.

8.8 Performance Evaluation Discussion

System performance was evaluated across multiple operational conditions

Parameter	Observation
Face Detection	High accuracy
Recognition Speed	Real-time capable
Multi-face Handling	Supported
Unknown Detection	Accurate
Low-light Performance	Moderate

Table 4: Performance Evaluation

8.9 Functional Outcome Analysis

The UVPAS system successfully achieved its intended objectives:

- Automated CCTV monitoring
- Identity recognition
- Dataset registration
- Real-time webcam tracking
- Unknown person detection
- Administrative dataset control

These outcomes confirm the feasibility of deploying AI-based surveillance in educational institutions.

8.10 Processing Time Analysis

Processing speed is crucial for surveillance usability.

Observed Performance

- Frame extraction → Fast
- Face detection → Real-time capable
- Recognition matching → Milliseconds per face
- Video processing → Efficient for institutional use

GPU acceleration further improves processing time where available.

CHAPTER – IX

CONCLUSION & FUTURE ENHANCEMENT

9.1 Conclusion

The development of the University Video Processing & Analysis System (UVPAS) marks a significant advancement in intelligent institutional surveillance and automated identity recognition. The primary objective of this project was to design and implement a smart CCTV monitoring system capable of detecting, identifying, and managing individuals through deep learning–based facial recognition.

The system successfully integrates video analytics, facial dataset management, and real-time recognition within a unified web-based platform. By leveraging computer vision techniques and the FaceNet deep learning model, the system demonstrates the ability to accurately extract facial embeddings and perform identity matching from both recorded CCTV footage and live webcam streams.

One of the key achievements of the project is the automation of surveillance monitoring. Traditional CCTV systems rely heavily on manual observation, which is time-consuming and prone to human error. The proposed system eliminates these limitations by automatically detecting faces, recognizing registered individuals, and flagging unknown persons. This significantly enhances monitoring efficiency and institutional security.

The implementation of multi-angle facial dataset registration (front, left, and right views) further improves recognition accuracy under pose variations. The integration of Excel-based dataset management simplifies bulk student data handling, making the system scalable for large institutional deployments.

From a performance perspective, the system demonstrated:

- High face detection accuracy
- Reliable identity recognition
- Real-time webcam processing capability
- Efficient video analytics
- Accurate unknown person detection

The modular architecture ensures flexibility, allowing additional intelligent surveillance features to be integrated without disrupting the existing workflow.

Overall, the UVPAS system proves to be a practical, cost-effective, and scalable AI-powered surveillance solution suitable for universities, hostels, offices, and secure campuses. It reduces manual monitoring effort while improving safety, accountability, and digital record management.

9.2 Future Enhancement

Although the developed system achieves its core objectives successfully, there are several opportunities for enhancement to further improve intelligence, scalability, and real-world applicability.

1. Real-Time Multi-Camera Integration

Future versions can support simultaneous monitoring from multiple CCTV cameras across campus. This would enable centralized surveillance control and large-area coverage.

2. Mobile Application Integration

Developing an Android/iOS mobile application would allow administrators to:

- Monitor live recognition feeds
- Receive intrusion alerts
- Access student recognition reports remotely

3. AI-Based Behavior Analysis

Beyond face recognition, the system can be extended to analyze suspicious activities such as:

- Loitering detection
- Crowd formation alerts
- Restricted area intrusion

This would transform the platform into a complete intelligent surveillance suite.

4. Masked Face Recognition

Enhancing the model to recognize partially occluded faces (masks, helmets, scarves) would improve performance in post-pandemic and security-sensitive environments.

5. Cloud-Based Deployment

Migrating the system to cloud infrastructure would provide:

- Scalable storage
- Remote accessibility
- High-performance GPU processing
- Centralized dataset management

6. Criminal / Blacklist Database Integration

Future enhancement can include integration with law enforcement or institutional blacklist databases to automatically flag restricted individuals.

7. Attendance & Access Control Automation

The system can be extended to support:

- Automated classroom attendance
- Hostel entry logging
- Lab access authentication

This would reduce manual record maintenance.

8. Gait Recognition Expansion

Since your system already includes gait gallery modules, future work can enhance walking pattern recognition to identify individuals even when faces are not visible.

9. Explainable AI Visualization

Adding heatmaps or attention maps would help visualize which facial regions influenced recognition decisions, improving transparency and trust.

REFERENCES

[1] Alhanaee, K., et al. (2021). Smart Attendance Tracking Using Convolutional Neural Networks and Deep Learning Techniques.

[2] Bugingo, E., et al. (2025). Improving Attendance Management through Face Recognition with Real-Time Facial Tracking.

[3] Sharma, P., & Sahu, A. (2022). Intelligent Attendance Management System Using Face Recognition and Machine Learning.

[4] DG HL, et al. (2020). Face Recognition based Attendance System.

[5] Wanjare, S. B., Mahajan, J. R., & Gadekar, S. S. (2024). Contactless Smart Attendance System Using Facial Recognition and QR Code.

[6] Li, H., et al. (2022Real-Time Student Attendance Monitoring Using Face Recognition with CNN and MTCNN.Procedia Computer Science, 199, 428–435.

[7] Mehmood, R., & Ullah, H. (2023). Smart Attendance Submission ID trn:oid:3618:116263561 Monitoring System Using Face Recognition with Liveness Verification. Sensors, 23(3), 1125. Submission ID trn: oid:::3618:116263561 Page 9 of 9 - Integrity Submission 3 2 12 2 7

[8] Rao, A. (2022). AttenFace: A Real-Time Student Attendance Management System Using Facial Recognition.

[9] Saraswat, D., et al. (2023). Contactless Attendance Monitoring System with Anti-Spoofing Mechanisms.

[10] Dixit, H. V. (2024). Automated Attendance System Using Face Recognition.

[11] Kulkarni, S. (2023). Attendance Monitoring System using Face Recognition.

[12] Budiman, A., et al. (2023). Student Attendance System Using Face Recognition with LBPH and CNN Techniques.

[13] Smart Attendance System: Using Facial Recognition 11 9 Technology. (2024).

[14] Masutage, A. A., Raynade, S. C., Patil, P. B., et al. (2023). Facial Recognition Attendance System (FRAS) based on Image Processing.

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Date of acceptance: 27.10.2025

Your submission has undergone review and the suggested corrections may be incorporated in the **camera-ready version** of your paper.

A detailed **camera-ready submission template will be shared with you before the final submission deadline**.

Review Comments (to be addressed in the camera-ready submission):

Reviewer 1

The paper presents a highly pertinent solution addressing the critical security and energy efficiency trade-off in resource-constrained IoT. The proposed cross-layer architecture's dual approach—integrating Machine Learning models (DT/LSTM) for robust anomaly detection and **Speck encryption** for energy savings—is both innovative and necessary. The documented **30% reduction in energy consumption** alongside high resilience against common threats (sinkhole, jamming) validates the architecture's practical feasibility. The discussion on future enhancements like Federated Learning (FL) and Adversarial Machine Learning (AML) demonstrates excellent foresight regarding privacy and robustness. Minor suggestions are needed to finalize the contribution: detailing the specific layer assignments for DT and LSTM, and providing more quantitative metrics (e.g., detection delay) to solidify the "high resilience" claim.

Recommendation: Accept with minor revisions.

Reviewer 2

This work delivers a meticulously engineered security framework for IoT that utilizes a sensible mix of established ML techniques and lightweight cryptography. The methodology is strengthened by validation across a **Cooja/Contiki OS simulation** and a **real-world Z1 motes testbed**, which significantly bolsters the practical relevance of the findings. The modular nature of the system applying different ML models for specific security tasks across layers is a strong point. To enhance the paper's overall impact, two small refinements are suggested: a brief, explicit justification for the selection of **Decision Trees and LSTMs**

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